# Fatigue 3D Model

## Statistical Analysis Tool

## User Manual

Version 1.0

**Program:** `Fatigue3DModel.html`

**Purpose:** Three-dimensional statistical modelling of fatigue data using hyperbolic regression

**Platform:** Any modern web browser

**Input format:** CSV (comma-separated values)

This document describes the complete workflow of the tool, from data loading to the generation of percentile hyperbola plots.

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# Introduction

**Fatigue 3D Model** is a self-contained, browser-based statistical analysis tool designed to fit a three-dimensional hyperbolic regression model to fatigue or similar engineering data. The program is distributed as a single HTML file (`Fatigue3DModel.html`) and requires no installation: it runs entirely inside any modern web browser.

## Purpose

The tool addresses the problem of characterising the statistical distribution of a *response variable* (e.g. fatigue life) as a function of two *predictor variables* (e.g. stress amplitude and a third physical quantity such as stress ratio, temperature, or frequency). By fitting a multivariate linear model in the transformed space $x \cdot y \cdot z$ and computing empirical percentiles of the residuals, the tool generates a family of hyperbolic iso-percentile curves that cover the full probability range from $P_0$ to $P_{99}$.

## Key Features

- Automatic detection and application of a base-10 logarithmic transformation for columns whose range spans more than three orders of magnitude.
- Interactive scatter plot of the raw data with click-to-toggle outlier identification.
- Ordinary least-squares (OLS) fitting of a seven-parameter bilinear model in the product space.
- Automatic outlier detection based on the asymptotes of the fitted hyperbola.
- Computation of residual quantiles at seven probability levels: $P_0$, $P_5$, $P_{25}$, $P_{50}$, $P_{75}$, $P_{95}$, and $P_{99}$.
- Generation of iso-percentile hyperbola plots for five representative values of the third variable, both in original scale and in the uniform $[0, 1]$ probability scale.
- One-click PDF export of every interactive plot.

## System Requirements

- A modern web browser (Chrome, Firefox, Edge, or Safari — version released after 2020 recommended).
- An internet connection is required on first use so that the browser can load the external libraries (Plotly, jsPDF, html2canvas) from their CDN addresses. Subsequent uses may work offline if the browser has cached those resources.
- No installation, no server, no database.

# Input Data Format

The program reads a **comma-separated values (CSV)** file. The file must satisfy the following requirements.

## File Structure

- **First row:** column headers, separated by commas.
- **Subsequent rows:** numeric data, one observation per row.
- **Missing values:** leave the cell empty or enter a non-numeric token; the program will treat it as `null` and skip that observation for any computation involving the missing column.
- **Encoding:** UTF-8 plain text.
- **Decimal separator:** period (`.`).

## Required Columns

The tool expects **exactly three numeric columns**. Their roles (horizontal variable $X$, vertical variable $Y$, and third variable $Z$) are assigned interactively after loading, so the physical order in the file is arbitrary.

| Column | Role | Typical example (fatigue) |
|---|---|---|
| 1st | $X$ — horizontal predictor | Stress amplitude $\sigma_a$ |
| 2nd | $Y$ — vertical response | Fatigue life $N$ |
| 3rd | $Z$ — third predictor | Stress ratio $R$ |

## Automatic Logarithmic Transformation

At load time the program inspects each column. If the ratio max / min $> 1000$ and all values are strictly positive, the entire column is replaced by $\log_{10}(\text{value})$ before any further computation. The column header is then labelled `log(OriginalName)` throughout the interface.

**Note**

The transformation is applied silently. After loading, check the information panel to confirm which columns have been log-transformed.

## Minimal Example

```
sigma_a,N,R
200,1.2e6,0.1
250,4.5e5,0.1
300,1.8e5,0.1
200,9.8e5,-0.5
250,3.7e5,-0.5
```

# Program Interface Overview

The interface is divided into six numbered sections that follow the natural analysis workflow. They are presented sequentially from top to bottom on the page.

| Part | Title | Purpose |
|---|---|---|
| I | Data Reading and Variable Definition | Load the CSV file and choose which column plays each role. |
| II | Data Table | Inspect the data in original or $[0, 1]$ scale; identify outliers visually. |
| III | Three-dimensional Model Fitting | Fit the OLS model and inspect residuals. |
| IV | (internal) | Outlier detection (executed as part of Part III). |
| V | Percentile Calculation | Display the seven residual quantiles. |
| VI | Hyperbolas for Different $Z$ Values | Generate and export the iso-percentile curves. |

# Step-by-Step Workflow

## Part I — Data Loading and Variable Definition

### Step 1 — Open the tool

Open `Fatigue3DModel.html` in your web browser by double-clicking the file (local mode) or by navigating to the hosted URL (GitHub Pages mode). The page detects its environment automatically.

### Step 2 — Set the bandwidth parameter

The **var3bandwidth** input (default `0.10`) controls the neighbourhood size used when computing local statistics for a given $Z$ value. A value of `0.10` means that data points within $\pm 10\%$ of the full $Z$-range around the target value are included. Acceptable values are strictly between 0 and 1.
Click **Apply var3bandwidth** to confirm the value. A green confirmation message will appear.

### Step 3 — Load the CSV file (local mode)

Click **Choose File**, select your `.csv` file, then click **Load File**. A success message confirms the number of rows and columns detected, together with a list of any log-transformed columns.

### Note

When the tool is served from GitHub Pages it automatically loads a file named `DataFatigue.csv` located in the repository root. In that case the file-picker control is hidden.

### Step 4 — Assign variables

Two drop-down menus appear after a successful load:

- **Vertical Variable (Y):** the response to be plotted on the vertical axis.

- **Horizontal Variable (X):** the predictor on the horizontal axis.

The remaining column is automatically designated as the **third variable** ($Z$). Click **Apply Selection**. An interactive scatter plot of $Y$ vs. $X$ is displayed immediately below.

**Warning**

You must select two *different* columns for $X$ and $Y$. The tool will alert you if the same column is chosen for both.

## Part II — Data Table

The data table shows all observations. Rows highlighted in red correspond to observations currently classified as outliers.

The toggle button **Show Data in [0,1] Scale** / **Show Original Data** switches between the original (possibly log-transformed) values and the empirical CDF-based probability scale. The $[0, 1]$ scale is computed column-by-column using linear interpolation of the empirical CDF:

$$u_i = \frac{\mathrm{rank}(x_i)}{n}, \qquad i = 1, \ldots, n.$$

## Part III — Three-dimensional Model Fitting

Click **Fit Model and Analyze**. The program executes the following sequence automatically:

### Model Specification

The tool fits a seven-parameter ordinary least-squares model. Define $x = X_i$, $y = Y_i$, $z = Z_i$ for the $i$-th observation. The model is linearised by expanding the product $xyz$ and expressing the response as:

$$-x_i y_i z_i \;=\; C_0 + C_1 x_i + C_2 y_i + C_3 z_i + C_{12}\, x_i y_i + C_{13}\, x_i z_i + C_{23}\, y_i z_i + \varepsilon_i$$

This can be written compactly as $\mathbf{y} = \mathbf{X}\boldsymbol{\theta} + \boldsymbol{\varepsilon}$, where the design matrix $\mathbf{X}$ has columns $[1,\, x,\, y,\, z,\, xy,\, xz,\, yz]$ and the parameter vector is $\boldsymbol{\theta} = [C_0, C_1, C_2, C_3, C_{12}, C_{13}, C_{23}]^\top$.

The OLS estimator $\hat{\boldsymbol{\theta}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y}$ is solved using Gaussian elimination with partial pivoting.

### Goodness-of-Fit Statistics

After fitting, the following statistics are displayed:

| Statistic | Definition |
|---|---|
| Mean of residuals | $\bar{\varepsilon} = \frac{1}{n}\sum_i \varepsilon_i$ |
| Std. deviation | $s = \sqrt{\frac{1}{n-1}\sum_i(\varepsilon_i - \bar{\varepsilon})^2}$ |
| Variance | $s^2$ |
| MSE | $\frac{1}{n}\sum_i \varepsilon_i^2$ |
| RMSE | $\sqrt{\text{MSE}}$ |
| MAE | $\frac{1}{n}\sum_i \lvert\varepsilon_i\rvert$ |
| $R^2$ | $1 - \text{SS}_{\text{res}}/\text{SS}_{\text{tot}}$ |
| MAPE | $\frac{100}{n}\sum_i \lvert\varepsilon_i/y_i\rvert$ (when $y_i \neq 0$) |

### Residual Empirical CDF Plot

A plot of the empirical cumulative distribution function of the residuals $F(\varepsilon)$ is generated. The seven percentile values are overlaid as coloured markers. The plot can be exported to PDF via the **Download Plot as PDF** button.

### Automatic Outlier Detection

Outlier detection is carried out in one pass over five equally spaced $Z$ reference values spanning the observed range $[Z_{\text{min},Z_{\text{max}}]}$. For each reference value $z_k$ the fitted model defines a rectangular hyperbola with:

- a **vertical asymptote** at $X = X_{\text{asym}}(z_k)$,
- a **horizontal asymptote** at $Y = Y_{\text{asym}}(z_k)$.

Any observation within the bandwidth of $z_k$ that lies to the *left* of the vertical asymptote or *below* the horizontal asymptote is flagged as an outlier and removed from subsequent computations. The scatter plot is refreshed and outlier rows are highlighted in the data table.

**Warning**

At least **7 valid observations** must remain after outlier removal for the analysis to continue. The program will halt and display a warning if this condition is not met.

## Part V — Percentile Calculation

Seven percentiles of the global residual distribution are computed and displayed:

$$P_0, \quad P_5, \quad P_{25}, \quad P_{50}, \quad P_{75}, \quad P_{95}, \quad P_{99}$$

These are obtained by linear interpolation between the order statistics of the sorted residual vector. The quantile at probability level $p$ is:

$$Q(p) = (1-w)\,\varepsilon_{(j)} + w\,\varepsilon_{(j+1)}, \qquad j = \lfloor p(n-1) \rfloor, \quad w = p(n-1) - j.$$

# Part VI — Hyperbola Plots

## Hyperbola Generation

For each of five reference values $z_k$ (equally spaced in $[Z_{\min,Z_{\max}}]$), the program computes seven iso-percentile hyperbolic curves. Given a quantile $q$ and fixed $z_k$, the $Y$ value corresponding to a given $X$ is obtained by solving the model equation for $Y$:

$$Y(X;\, z_k,\, q) = \frac{q - C_0 - C_1 X - C_3 z_k - C_{13} X z_k}{C_2 + C_{12} X + C_{23} z_k}$$

where $q$ replaces the residual $\varepsilon$ term. Only points that lie in the physically valid region (beyond both asymptotes) are plotted.

## Original-Scale Plots

For each reference $Z$ value a two-panel row is displayed:

- **Left panel:** the seven iso-percentile hyperbolas overlaid on the scatter plot. Data points whose $Z$ value falls within the bandwidth are shown in orange; outliers in blue; remaining data in light grey.
- **Right panel:** the empirical CDF of the local residuals (those belonging to the bandwidth neighbourhood of $z_k$), with the seven percentile markers.

A line of text below each row lists the local percentile values for that $Z$ slice.

## Plots in $[0, 1]$ Probability Scale

Click **Generate Plots in [0,1] Scale** to produce an additional set of plots where both axes are transformed to the uniform probability scale using the empirical CDF:

$$u_X = F_X(X), \qquad u_Y = F_Y(Y).$$

In this representation the axes run from 0 to 1 and all variables are treated symmetrically, facilitating visual comparison across different datasets.

## Colour Convention

All hyperbola plots use the following consistent colour scheme:

| Percentile | Colour | Line style |
|---|---|---|
| $P_0$ | Red (`#FF0000`) | Solid |
| $P_5$ | Orange-red (`#FF4500`) | Solid |
| $P_{25}$ | Dark orange (`#FF8C00`) | Solid |
| $P_{50}$ | Orange (`#FFA500`) | **Dashed** |
| $P_{75}$ | Steel blue (`#4682B4`) | Solid |
| $P_{95}$ | Dodger blue (`#1E90FF`) | Solid |
| $P_{99}$ | Royal blue (`#4169E1`) | Solid |

# Exporting Results

Every plot generated by the tool can be exported individually as a PDF file. Clicking any **Download Plot as PDF** button (shown in red) performs the following sequence internally:

1. The plot is temporarily resized to $1200 \times 900$ pixels with a white background.
2. A PNG image is captured at $2\times$ pixel density ($2400 \times 1800$ px).
3. The image is embedded in a landscape A4 PDF page using the jsPDF library.
4. The PDF is downloaded through the browser under an automatically generated filename (e.g. `Hyperbola_0.250.pdf` or `Hyperbola_01_0.250.pdf` for the $[0, 1]$-scale version).
5. The plot is restored to its original display size.

# Mathematical Background

## The Three-Variable Hyperbolic Model

The fundamental relationship exploited by the tool is that, for fixed $z$, the implicit equation

$$C_0 + C_1 x + C_2 y + C_3 z + C_{12}\, xy + C_{13}\, xz + C_{23}\, yz \;=\; -\,xyz$$

defines a rectangular hyperbola in the $(x, y)$ plane. Rearranging:

$$\begin{aligned} y\,(C_2 + C_{12}\, x + C_{23}\, z) \;&=\; -xyz - C_0 - C_1 x - C_3 z - C_{13}\, xz \\ y \;&=\; \frac{-xyz - C_0 - C_1 x - C_3 z - C_{13}\, xz}{C_2 + C_{12}\, x + C_{23}\, z}. \end{aligned}$$

The **vertical asymptote** is the value of $x$ for which the denominator (or an equivalent grouping) becomes zero, and the **horizontal asymptote** follows symmetrically.

## Iso-Percentile Curves

Adding a residual quantile $q$ to the right-hand side shifts the hyperbola vertically in the product space:

$$C_0 + C_1 x + C_2 y + C_3 z + C_{12}\, xy + C_{13}\, xz + C_{23}\, yz + q \;=\; -\,xyz.$$

Each value of $q$ (i.e. each of the seven percentile levels) produces a distinct hyperbola. Together they form the *iso-percentile fan* displayed in Part VI.

### Empirical CDF and $[0, 1]$ Scale

The empirical cumulative distribution function is estimated by the **Weibull plotting position**:

$$\hat{F}(x_{(i)}) = \frac{i}{n}, \qquad i = 1, \ldots, n,$$

where $x_{(1)} \leq x_{(2)} \leq \cdots \leq x_{(n)}$ are the order statistics. Linear interpolation is used between adjacent pairs to map any value onto $[0, 1]$ and, conversely, to invert from $[0, 1]$ back to the original scale.

## Troubleshooting

| Symptom | Likely cause and remedy |
|---|---|
| *"Plotly has not loaded"* red banner on startup | The browser could not reach the Plotly CDN. Check your internet connection and reload the page. |
| *"Please select a CSV file"* alert when clicking Load | No file was selected in the file picker. Click the picker first. |
| File loads but columns are wrong / missing | Verify that the CSV uses commas as delimiters, that the first row contains column names, and that all data values use a period as the decimal separator. |
| *"Not enough valid data for fitting (minimum 7 points)"* | The dataset has fewer than 7 rows with complete (non-null, finite) values in all three columns. Check for missing values or remove rows with `NaN`. |
| *"Singular matrix"* error during fitting | Two or more columns (after log-transformation) are linearly dependent. This can happen if one variable is a constant or a perfect linear function of another. |
| Hyperbola curves appear truncated or absent | The curves may fall entirely outside the visible axis range. Try a different variable assignment or verify that the bandwidth parameter captures enough local data points. |
| PDF download button is greyed out | The corresponding plot has not been generated yet. Complete the preceding analysis step first. |
| *"No valid data for plot"* message in scatter area | All rows contain at least one null or non-finite value in the selected variables. Inspect the raw CSV for formatting issues. |

## Parameter Reference

| Parameter | Location | Description |
|---|---|---|
| `var3bandwidth` | Part I control panel | Fractional half-width of the $Z$-neighbourhood used for local statistics. Range: $(0, 1)$. Default: `0.10`. |
| $C_0, \ldots, C_{23}$ | Displayed in Part III after fitting | Seven OLS regression coefficients of the linearised hyperbolic model. |
| $P_0, P_5, P_{25}, P_{50}, P_{75}, P_{95}, P_{99}$ | Displayed in Parts III, V, VI | Empirical quantiles of the global residual distribution. |
| `ecdfX`, `ecdfY` | Internal (used in Part VI) | Empirical CDF lookup tables for the $X$ and $Y$ columns, used for the $[0, 1]$-scale transformation. |

# Quick-Start Checklist

1. Open `Fatigue3DModel.html` in your browser.
2. Confirm the green *"Plotly loaded successfully"* banner.
3. (Optional) Adjust `var3bandwidth` and click **Apply var3bandwidth**.
4. Select your CSV file and click **Load File**.
5. Verify the column list and log-transformation notice.
6. Select the **Vertical** and **Horizontal** variables; click **Apply Selection**.
7. Inspect the scatter plot; use **Show Data in [0,1] Scale** if desired.
8. Click **Fit Model and Analyze**.
9. Read the parameter estimates and goodness-of-fit metrics in Part III.
10. Acknowledge the outlier-detection alert (lists removed points).
11. Inspect the residual CDF plot and the $P$-values in Part V.
12. Scroll to Part VI to view the iso-percentile hyperbola plots.
13. Click **Generate Plots in [0,1] Scale** for the normalised view.
14. Use the red **Download Plot as PDF** buttons to export any plot.

# Glossary

**Bandwidth** (`var3bandwidth`)
: The half-width of the sliding window in the $Z$-direction, expressed as a fraction of the total $Z$-range.

**ECDF**
Empirical Cumulative Distribution Function. A step function that estimates the true CDF from the sample.

**Horizontal asymptote**
The value $Y_{\text{asym}}$ towards which the fitted hyperbola converges as $X \to \infty$.

**Iso-percentile curve**
A hyperbola in the $(X, Y)$ plane along which the probability that a random observation falls below the curve is constant.

**Log-transformation**
Replacement of raw values by their base-10 logarithm, applied automatically to columns with a dynamic range exceeding three orders of magnitude.

**MAE**
Mean Absolute Error: $\frac{1}{n} \sum |\varepsilon_i|$.

**MAPE**
Mean Absolute Percentage Error: $\frac{100}{n} \sum |\varepsilon_i / y_i|$.

**MSE / RMSE**
Mean Squared Error / Root Mean Squared Error.

**OLS**
Ordinary Least Squares regression.

**Outlier**
A data point that lies on the wrong side of a fitted hyperbola's asymptote, inconsistent with the physical model.

**Percentile $P_k$**
The value below which $k\%$ of the empirical residual distribution falls.

**Residual $\varepsilon_i$**
Difference between the model prediction and the observed response for observation $i$ in the transformed product space.

**Third variable ($Z$)**
The conditioning variable whose value determines which hyperbola slice is displayed. Typical examples: stress ratio, temperature, frequency.

**Vertical asymptote**
The value $X_{\text{asym}}$ at which the fitted hyperbola becomes undefined (denominator $\to 0$).